

ALSHUMOOKH GROUP

Settlement Gateway v2.0 — Technical Integration Division

TECHNICAL INTEGRATION ROADMAP & API SPECIFICATION

HIPERCAPITAL FINANCE ↔ ALSHUMOOKH SETTLEMENT GATEWAY v2.0

VERSION 4.0 — COMPLETE TECHNICAL SPECIFICATION

Prepared by:	ALSHUMOOKH GROUP — Technical Operations
Document Type:	External Integration Specification (Full API Reference)
Classification:	CONFIDENTIAL — For HIPERCAPITAL FINANCE Only
Date:	08 June 2026
Reference:	ASIG-HIPC-TECH-2026-v4
Gateway Version:	Alshumookh Settlement Gateway v2.0 (FastAPI)

■ **IMPORTANT:**

Proprietary API specifications of ALSHUMOOKH GROUP settlement infrastructure.
All integration requests are logged, monitored and audited. Unauthorized distribution is prohibited.

TABLE OF CONTENTS

- 1. Executive Summary & Integration Overview
- 2. System Architecture & Security Pipeline
- 3. Authentication Methods — Full Specification
- 4. Main Ingest Endpoint — POST /api/v1/payloads/ingest
- 5. Complete JSON Payload Schema & Field Aliases
- 6. Payload & Transfer Status Codes (All States)
- 7. All Error Codes & Rejection Reasons
- 8. Complete API Endpoints Reference
- 9. Blockchain Networks, Contracts & Receiver Addresses
- 10. Webhook & Callback Specification
- 11. M1 Fund / Wire Transfer Pathway
- 12. Integration Phases & Compliance Requirements
- 13. Live cURL & Request Examples
- 14. Go-Live Compliance Checklist

1. EXECUTIVE SUMMARY & INTEGRATION OVERVIEW

ALSHUMOOKH GROUP operates the Alshumookh Settlement Gateway v2.0 (ASIG), a production FastAPI blockchain settlement platform. This document provides HIPERCAPITAL FINANCE with every technical detail required to integrate their HIPCF Chain (Chain ID 240884) with ASIG — all endpoints, field aliases, status codes, error codes, and request examples are drawn directly from the live codebase.

Component	ASIG (Production)	HIPCF Requirement
Gateway	Alshumookh Settlement Gateway v2.0	REST API integration over HTTPS
Framework	FastAPI · Python 3.11+	Standard JSON/HTTPS client
Accepted Assets	USDT (ERC-20/TRC-20) · USDC · M1 Token	HCEURO must bridge to USDT first
Supported Networks	Ethereum · Base · Tron	HIPCF Chain NOT natively supported
Blockchain Verification	Alchemy Webhooks + EVM RPC	Public chain tx_hash mandatory
Auth Methods	API Key · OAuth2 · HMAC · JWS · JWE · mTLS	Min: API Key + HMAC-SHA256
Rate Limit	240 req/60s global · 200 req/60s per key	Must implement request queuing
Confirmation Threshold	6+ blocks (ETH/Base) · 20 blocks (Tron)	Wait before marking settled

x BACKLOG: ASIG does NOT accept asset=TOKENIZED_USD, asset=USDX, asset=HCEURO, or settlement_type=PRIVATE_INTERNAL_LEDGER. HIPERCAPITAL must bridge HCEURO to USDT on Ethereum or Base before any payload submission.

2. SYSTEM ARCHITECTURE & SECURITY PIPELINE

■ REQUEST PIPELINE: IP Allowlist → API Key / OAuth2 Auth → Idempotency-Key Check → mTLS Certificate (optional) → JWE Decrypt (optional) → Timestamp Validation → JWS Verify (optional) → HMAC-SHA256 Validate → JSON Normalize + Alias Resolve → Database Persist → Alchemy Blockchain Verify → Confirmation → Reconcile

Security Layer	Header / Mechanism	Required?	Notes
IP Allowlist	Server-side IP check	✓ Required	Egress IPs registered in ASIG dashboard
API Key	X-API-Key:	✓ Required	Generated from ALSHUMOOKH admin panel
Idempotency Key	Idempotency-Key:	✓ Required	Unique UUID per request — deduplication
OAuth2 Bearer	Authorization: Bearer	Optional	Required if client set require_oauth=true
HMAC-SHA256	X-HMAC-Signature:	Recommended	Covers method + url + timestamp + body_hash
Timestamp	X-Timestamp:	With HMAC	Max 5 minute drift from server time (UTC)
JWS	X-JWS-Signature:	Optional	JSON Web Signature of full request body
JWE	Content-Type: application/jose	Optional	Full body as JWE compact serialization
mTLS	X-Client-Cert-Fingerprint:	Optional	Client cert SHA-256 fingerprint validation

3. AUTHENTICATION METHODS — FULL SPECIFICATION

3.1 API Key — Minimum Required

Every request must include X-API-Key. Keys are generated from the ALSHUMOOKH admin dashboard. Invalid/inactive key → HTTP 401 error_code: invalid_api_key.

Headers:

```
X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000
```

3.2 OAuth2 Client Credentials

POST to /api/v1/oauth/token with client_id + client_secret. Token expires in 3600s. Required when client config sets require_oauth=true.

Token Request + Response:

```
POST /api/v1/oauth/token
Content-Type: application/x-www-form-urlencoded
grant_type=client_credentials&client_id=YOUR_ID&client_secret=YOUR_SECRET

{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "token_type": "bearer", "expires_in": 3600 }
```

Usage:

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

3.3 HMAC-SHA256 Request Signing — Recommended

Signature covers: METHOD + URL + Unix_timestamp + SHA256(body). Add X-HMAC-Signature and X-Timestamp headers. Max 5 min clock drift.

Python construction:

```
import hmac, hashlib, time, json
body = json.dumps(payload)
timestamp = str(int(time.time()))
body_hash = hashlib.sha256(body.encode()).hexdigest()
message = f'POST\n/api/v1/payloads/ingest\n{timestamp}\n{body_hash}'
signature = hmac.new(secret.encode(), message.encode(), hashlib.sha256).hexdigest()
# Headers:
X-HMAC-Signature: <signature_hex>
X-Timestamp: <unix_timestamp>
```

4. MAIN INGEST ENDPOINT — POST /api/v1/payloads/ingest

Property	Value
Endpoint	POST /api/v1/payloads/ingest
Content-Type	application/json (or application/jose for JWE-encrypted payload)
Required Headers	X-API-Key + Idempotency-Key:
Optional Headers	X-HMAC-Signature + X-Timestamp Authorization: Bearer X-JWS-Signature
Rate Limit	200 req / 60s per API key 240 req / 60s global
Success (with tx_hash)	HTTP 200 · status: payload_received · verification_status: ALCHEMY_PENDING
Success (no tx_hash)	HTTP 200 · status: payload_received · verification_status: AWAITING_TX_HASH
Parse Failed	HTTP 200 · status: payload_received_unparsed · verification_status: RECEIVED
Idempotency Replay	HTTP 200 · status: payload_received (replayed) · parsed: true/false
Auth Failure	HTTP 401/403 · JSON error body with error_code field – see Section 7
Validation Failure	HTTP 400 · missing required field or malformed request

4.1 Full cURL Example

```
curl -X POST https://api.alshumookh.com/api/v1/payloads/ingest \
-H 'Content-Type: application/json' \
-H 'X-API-Key: asig_live_XXXXXXXXXXXXXXXXXXXX' \
-H 'Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000' \
-H 'X-HMAC-Signature: 3a7f9c2e4b8d1f...' \
-H 'X-Timestamp: 1749461234' \
-d '{"transaction_reference":"HIPC-2026-001","tx_hash":"0xabc123...",
"sender_wallet":"0xYOUR_WALLET","receiver_wallet":"0xBD682cfd8382a90adfDd6745780D3D7959c4d939",
"amount":"50000.00","asset":"USDT","network":"ethereum",
"token_contract":"0xdAC17F958D2ee523a2206206994597C13D831ec7",
"callback_url":"https://your-server.com/webhook/asig"}'
```

4.2 Response Examples

Scenario	HTTP	Response JSON (key fields)
With tx_hash (normal)	200	status: "payload_received" · verification_status: "ALCHEMY_PENDING" · payload_id:
Without tx_hash	200	status: "payload_received" · verification_status: "AWAITING_TX_HASH"
Parse failed	200	status: "payload_received_unparsed" · verification_status: "RECEIVED"
Idempotency replay	200	status: "payload_received" · parsed: true · (original response replayed)
Invalid API key	401	{"error": "invalid_api_key", "message": "API key is invalid or inactive"}
Missing Idempotency-Key	400	{"error": "missing_idempotency_key", "message": "Idempotency-Key header is required"}
IP not allowed	403	{"error": "ip_not_allowed", "message": "Client IP is not allowed"}

5. COMPLETE JSON PAYLOAD SCHEMA & FIELD ALIASES

ASIG's normalization engine auto-resolves incoming JSON field names using a comprehensive alias map. HIPERCAPITAL may use their own naming convention — the engine finds the first matching alias at the top level or one level deep within nested objects.

Canonical Field	Accepted Aliases	Type	Required
transaction_reference	transaction_id · txRef · reference · ref · transactionCode · referenceId · payment_reference	string	Recommended
tx_hash	transaction_hash · hash · blockchain_hash · txid · txHash · transactionHash	string (0x...)	Conditional
sender_wallet	from_wallet · from · source_wallet · fromAddress · from_address · senderAddress · origin_wallet	string (0x.../T...)	Required
receiver_wallet	to_wallet · to · destination_wallet · toAddress · to_address · recipientAddress · beneficiary_wallet	string (0x.../T...)	Required
amount	value · token_amount · transfer_amount · settlement_amount · tokenAmount · transferAmount	decimal string	Required
asset	currency · token · symbol · crypto_currency · tokenSymbol · cryptoCurrency	string	Required
network	chain · blockchain · protocol · chainName · networkName	string	Required
token_contract	contract_address · contract · token_address · contractAddress · tokenContract	string (0x...)	Recommended
timestamp	created_at · time · tx_time · txTime · transaction_time	ISO8601 / Unix	Optional
sender	sender_name · from_name · senderName · fromName · payer	string	Optional
receiver	receiver_name · to_name · beneficiary · receiverName · toName · payee	string	Optional
settlement_type	settlement · type · settlementType · paymentType · payment_type	string	Optional
authorization_code	auth_code · authorization · authCode · authorizationCode	string	Optional
callback_url	callback · webhook_url · notify_url · callbackUrl · webhookUrl · notifyUrl	URL	Optional
signature	sig · x_signature · xSignature · payloadSignature	hex	Optional
payload_hash	hash_payload · content_hash · payloadHash · contentHash	SHA-256 hex	Optional

5.1 Complete JSON Example (all fields)

```
{
  "transaction_reference": "HIPC-2026-001",
  "tx_hash": "0xabc123...def456",
  "sender_wallet": "0xYOUR_SOURCE_WALLET",
  "receiver_wallet": "0xBD682cfD8382a90adFd6745780D3D7959c4d939",
  "amount": "50000.00",
  "asset": "USDT",
  "network": "ethereum",
  "token_contract": "0xdAC17F958D2ee523a2206206994597C13D831ec7",
  "settlement_type": "wallet_to_wallet",
  "timestamp": "2026-06-09T10:00:00Z",
  "sender": "HIPERCAPITAL FINANCE",
  "receiver": "ALSHUMOOKH GROUP",
  "authorization_code": "HIPC-AUTH-2026",
  "callback_url": "https://your-server.com/webhook/asig"
}
```

5.2 HIPERCAPITAL Alternative Field Names (auto-resolved by ASIG)

```
{ "txRef": "HIPC-2026-001", "txHash": "0xabc123...", "fromAddress": "0xYOUR_WALLET",
  "toAddress": "0xBD682c...", "tokenAmount": "50000.00", "tokenSymbol": "USDT",
  "chainName": "ethereum", "contractAddress": "0xdAC17F...", "webhookUrl": "https://..." }
```

6. PAYLOAD & TRANSFER STATUS CODES (ALL STATES)

6.1 PayloadVerificationStatus — Inbound Settlement Flow

Status	Meaning	Trigger	Action Required
RECEIVED	Stored but parsing failed / unrecognized structure	Ingest (parse fail)	Check your JSON structure — contact ASIG ops
PARSED	Fields normalized and extracted successfully	Ingest (parse ok, no tx_hash)	Submit tx_hash when available
AWAITING_TX_HASH	Parsed but no blockchain tx_hash provided	Ingest without tx_hash	PATCH payload with tx_hash once broadcast
ALCHEMY_PENDING	tx_hash submitted — awaiting Alchemy confirmation	Alchemy webhook received	None — wait for confirmations
ALCHEMY_VERIFIED	Confirmed on-chain (6+ ETH/Base, 20 TRON blocks)	Alchemy confirms blocks	System auto-advances
ON_CHAIN_CONFIRMED	Transaction fully confirmed on blockchain	Block threshold met	ASIG reconciles — no action
RECONCILED	Settlement complete — funds reconciled ✓	Admin reconciliation	Process complete
MANUAL_REVIEW	Flagged: RPC error / TRON / amount mismatch	Verification anomaly	Contact ASIG ops with payload_id
FAILED	Blockchain verification failed or amount mismatch ✗	Verification failed	Resubmit with corrections or contact ops

6.2 OutboundTransferStatus — ASIG → HIPERCAPITAL Payments

Status	Meaning	Notes
PENDING	Transfer created, awaiting processing	Initial state on creation
AWAITING_APPROVAL	Requires admin approval before broadcast	Large transfers trigger approval workflow
APPROVED	Admin approved — queued for blockchain broadcast	Pre-broadcast state
BROADCASTING	Transaction being submitted to network	Broadcast in progress
COMPLETED	Confirmed on-chain — settlement complete ✓	Final success state
FAILED	Broadcast failed — retry or manual intervention ✗	Check error in transfer record
CANCELLED	Cancelled before broadcast	No funds moved

6.3 OrderStatus — Payment Orders (Coinbase / MoonPay / Circle)

Status	Meaning
CREATED	Order record created — awaiting provider confirmation
PENDING	Submitted to provider — payment pending
PROCESSING	Provider processing the payment
COMPLETED	Payment confirmed and settled ✓
FAILED	Payment failed — contact provider ✗
REFUNDED	Payment refunded to customer
EXPIRED	Order expired before payment received

7. ALL ERROR CODES & REJECTION REASONS

All ASIG error responses use a consistent JSON body. The `error_code` field uniquely identifies the rejection. Every rejection is also logged to the audit trail with the `event_type` below.

Error response structure:

```
{ "error": "<error_code>", "message": "Human-readable explanation" }
```

HTTP	error_code	Audit Event	Cause	Fix
401	missing_credentials	PAYLOAD_REJECTED_AUTH	No X-API-Key and no Authorization header	Add X-API-Key or Bearer token
401	invalid_api_key	PAYLOAD_REJECTED_AUTH	API key not found or inactive	Verify key from ALSHUMOOKH dashboard
401	oauth_required	PAYLOAD_REJECTED_AUTH	Client requires OAuth2 but API Key sent	GET token from /api/v1/oauth/token first
403	ip_not_allowed	PAYLOAD_REJECTED_AUTH	Caller IP not in client IP allowlist	Register your egress IP in ASIG dashboard
400	missing_idempotency_key	PAYLOAD_REJECTED_VALIDATION	Idempotency-Key header missing	Add Idempotency-Key: to every call
401	mtls_required	PAYLOAD_REJECTED_AUTH	Client requires mTLS — no cert fingerprint	Add X-Client-Cert-Fingerprint header
400	invalid_jwe	PAYLOAD_REJECTED_VALIDATION	JWE-encrypted body failed decryption	Check JWE key pair and algorithm
401	timestamp_expired	PAYLOAD_REJECTED_AUTH	X-Timestamp older than 5 minutes	Sync server clock to UTC (NTP)
401	invalid_jws	PAYLOAD_REJECTED_AUTH	JWS signature verification failed	Check signing key and JWS algorithm
401	hmac_required	PAYLOAD_REJECTED_AUTH	Client requires HMAC — none provided	Add X-HMAC-Signature + X-Timestamp
401	invalid_hmac	PAYLOAD_REJECTED_AUTH	HMAC-SHA256 verification failed	Check secret and message construction
400	invalid_json	—	Request body is not valid JSON	Set Content-Type: application/json
400	invalid_payload	—	Body is not a JSON object (e.g. array)	Body must be { } object
404	not_found	—	payload_id does not exist	Use payload_id returned from /ingest
429	rate_limit_exceeded	—	Exceeded 200/60s per key or 240/60s global	Implement backoff + request queue

8. COMPLETE API ENDPOINTS REFERENCE

8.1 Settlement & Payload Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/payloads/ingest	API Key + Idempotency	Submit inbound settlement payload — main integration endpoint
GET	/api/v1/payloads/schema	API Key	Retrieve canonical schema + supported field aliases
POST	/api/v1/payloads/m1-fund	API Key	M1 Fund inbound settlement (EUR wire / IBAN pathway)
GET	/api/v1/payloads/{id}	API Key	Get payload status and details by payload_id
POST	/api/v1/oauth/token	Client credentials	Obtain OAuth2 Bearer token (expires 3600s)

8.2 Payment Order Endpoints

Method	Endpoint	Auth	Description
POST	/api/v1/payments/orders	API Key	Create payment order (Coinbase/MoonPay/Circle/Stripe)
GET	/api/v1/payments/orders	API Key	List all payment orders
GET	/api/v1/payments/orders/{id}	API Key	Get payment order by ID
GET	/api/v1/payments/orders/{id}/status	API Key	Get real-time order status
GET	/api/v1/payments/widget-url	API Key	Get onramp widget URL
GET	/api/v1/payments/coinbase/onramp-url	API Key	Get Coinbase onramp URL
GET	/api/v1/payments/transactions	API Key	List all transactions
GET	/api/v1/payments/transactions/{id}	API Key	Get transaction detail by ID

8.3 Webhook Receivers (Blockchain & Payment Events)

Method	Endpoint	Auth	Description
POST	/webhooks/alchemy	Alchemy token	Blockchain tx confirmations — auto-advances payload status
POST	/webhooks/circle	Circle HMAC	Circle payment confirmations
POST	/webhooks/moonpay	MoonPay sig	MoonPay payment events
POST	/webhooks/coinbase	Coinbase sig	Coinbase Commerce payment events

8.4 Admin Endpoints (ASIG Operations / Authorized Partners)

Method	Endpoint	Auth	Description
GET	/api/v1/admin/payloads	Admin key	List payloads with filters (status, date, network, client)
GET	/api/v1/admin/payloads/{id}	Admin key	Full payload detail including audit trail
POST	/api/v1/admin/payloads/{id}/verify	Admin key	Trigger manual blockchain re-verification
PATCH	/api/v1/admin/payloads/{id}/review	Admin key	Update manual review decision
POST	/api/v1/admin/outbound-transfers	Admin key	Initiate USDT outbound transfer (ETH/TRON/Base)
GET	/api/v1/admin/outbound-transfers	Admin key	List outbound transfers with status
POST	/api/v1/admin/clients	Admin key	Create new API client
GET	/api/v1/admin/clients	Admin key	List all API clients
GET	/api/v1/admin/dashboard/stats	Admin key	System statistics and payload counts
GET	/api/v1/admin/treasury/balance	Admin key	Treasury wallet balances across all networks
POST	/api/v1/admin/fnfcu	Admin key	FNFCU transfer request submission

9. BLOCKCHAIN NETWORKS, CONTRACTS & RECEIVER ADDRESSES

9.1 Supported Token Contracts

Token	Network	Contract Address	Standard	Status
USDT	Ethereum Mainnet	0xdAC17F958D2ee523a2206206994597C13D831ec7	ERC-20	✓ Active
USDT	Tron	TR7NHqjeKQxGTCi8q8ZY4pL8otSzgjLj6t	TRC-20	■ Placeholder
USDC	Ethereum Mainnet	0xa0b86991c6218b36c1d19d4a2e9eb0ce3606eb48	ERC-20	✓ Active
USDC	Base Network	0x833589fcd6edb6e08f4c7c32d4f71b54bda02913	ERC-20 (Base)	✓ Active
M1 Token	Sepolia Testnet	0xD999DB972BBDc2C13a9595A1474A04F5e59169a5	ERC-20	Testnet
SIG Token	Sepolia Testnet	0x14E068D3C99cbC8BAe723D83D1761EF4C124A0ab	ERC-20	Testnet

9.2 ASIG Receiver Wallets & Bank Details

Channel	Address / Account	Asset Accepted
Ethereum / Base (EVM)	0xBD682cfD8382a90adfDd6745780D3D7959c4d939	USDT ERC-20 · USDC
Tron (TRC-20)	Contact ASIG ops for Tron address	USDT TRC-20
Wire Transfer IBAN	AE960260001015754367301	USD / EUR fiat
Bank	Emirates NBD – Sheikh Zayed Road, Dubai, UAE	—
BIC / SWIFT	EBILAEADXXX	—

9.3 Confirmation Thresholds

Network	Chain ID	Min Blocks	Avg Time	Verification
Ethereum Mainnet	1	6+ blocks	~90 seconds	Alchemy Webhook + EVM RPC
Base Network	8453	6+ blocks	~18 seconds	Alchemy Webhook + EVM RPC
Tron	N/A	20 blocks	~60 seconds	MANUAL_REVIEW (placeholder)
Sepolia Testnet	11155111	6+ blocks	~90 seconds	Alchemy (testing only)

10. WEBHOOK & CALLBACK SPECIFICATION

When payload verification_status changes, ASIG POSTs a JSON notification to the callback_url provided in the original payload. HIPERCAPITAL must deploy a public HTTPS endpoint to receive these updates. Endpoint must respond HTTP 200 within 10 seconds. ASIG retries 3× with exponential backoff.

10.1 Outbound Callback Payload

```
POST https://your-server.com/webhook/asig
Content-Type: application/json
X-ASIG-Signature: <hmac_sha256_hex>

{ "event": "payload_status_update",
  "payload_id": "a8f2c3d4-1234-5678-abcd-ef0123456789",
  "transaction_reference": "HIPC-2026-001",
  "verification_status": "ON_CHAIN_CONFIRMED",
  "previous_status": "ALCHEMY_VERIFIED",
  "tx_hash": "0xabc123...def456",
  "network": "ethereum", "amount": "50000.00", "asset": "USDT",
  "timestamp": "2026-06-09T10:05:22Z" }
```

10.2 Signature Verification

```
# Verify ASIG callback authenticity:
import hmac, hashlib
sig = request.headers['X-ASIG-Signature']
expected = hmac.new(webhook_secret.encode(), request.body, hashlib.sha256).hexdigest()
assert hmac.compare_digest(sig, expected) # reject if mismatch
```

11. M1 FUND / WIRE TRANSFER PATHWAY

For EUR-denominated or fiat wire settlements, use the M1 Fund inbound pathway. ASIG converts incoming EUR wire transfers via FX → USDT → M1 Token (ERC-20 on Sepolia).

Field	Value
IBAN	AE960260001015754367301
BIC / SWIFT	EBILAEADXXX
Bank	Emirates NBD – Sheikh Zayed Road, Dubai, UAE
M1 Ingest Endpoint	POST /api/v1/payloads/m1-fund
M1 Receiver ETH Wallet	0xBD682cFD8382a90adFDd6745780D3D7959c4d939
M1 Token Contract (Sepolia)	0xD999DB972BBdc2C13a9595A1474A04F5e59169a5
Tokenization Flow	EUR Wire → FX Conversion → USDT → M1 Token (Sepolia ERC-20)
Settlement Type Value	M1_FUND_INBOUND (auto-set by /m1-fund endpoint)

11.1 M1 Fund Request Example

```
POST /api/v1/payloads/m1-fund
X-API-Key: asig_live_xxxx
Content-Type: application/json

{ "grouped_m1_detail": { "total_group_value_eur": "250000.00",
"group_reference": "HIPC-M1-2026-001", "settlement_date": "2026-06-09" },
"sender": "HIPERCAPITAL FINANCE S.L.", "receiver": "ALSHUMOOKH GROUP",
"callback_url": "https://your-server.com/webhook/asig" }
```

12. INTEGRATION PHASES & COMPLIANCE REQUIREMENTS

PHASE 1 — Legal & Regulatory (Weeks 1–4)

Requirement	Detail
FCA Authorization	Obtain FCA auth for UK payment activities or appoint FCA-authorized partner
Reserve Audit	Third-party audit of 6.98 trillion EUR reserve claim (Big 4 or equivalent)
AML/KYC Docs	Provide AML policy + KYC program docs — ASIG requires FATF-compliant counterparties
Corporate Verification	UK Companies House cert + Spain CNMV registration + Panama incorporation

PHASE 2 — Technical Setup (Weeks 5–8)

Requirement	Detail
USDT Bridge	Deploy HCEURO → USDT bridge on Ethereum or Base (no external HCEURO liquidity today)
API Client Registration	Contact ASIG ops — receive API key + HMAC secret + client ID
IP Registration	Provide production egress IP range for ASIG firewall allowlist
Webhook Endpoint	Deploy HTTPS endpoint for ASIG status callbacks (see Section 10)
HMAC Implementation	Implement HMAC-SHA256 signing per Section 3.3 specification

PHASE 3 — Sepolia Testnet (Weeks 9–12)

Requirement	Detail
Testnet Credentials	Request Sepolia testnet API key — all tests on Chain ID 11155111
10 Test Payloads	Submit 10 payloads with real Sepolia tx_hash values
Status Flow Test	Confirm ALCHEMY_PENDING → ALCHEMY_VERIFIED → ON_CHAIN_CONFIRMED
Callback Test	Confirm webhook delivery for every status transition
Error Handling Test	Intentionally trigger each error_code to verify client error handling

PHASE 4 — Production Go-Live (Weeks 13–16)

Requirement	Detail
Security Review	ASIG security team reviews integration architecture
Load Test	Demonstrate 200 req/60s handling without rate limit violations
First Production Payment	Submit live USDT payment with real on-chain tx_hash
Reconciliation	ASIG ops confirms first RECONCILED status payload
SLA Sign-off	Sign ASIG integration SLA — 99.9% uptime, 48h support SLA

13. LIVE cURL & REQUEST EXAMPLES

13.1 Get OAuth2 Token

```
curl -X POST https://api.alshumookh.com/api/v1/oauth/token \
-H 'Content-Type: application/x-www-form-urlencoded' \
-d 'grant_type=client_credentials&client_id=YOUR_ID&client_secret=YOUR_SECRET'
```

13.2 Submit Payload (API Key + HMAC)

```
curl -X POST https://api.alshumookh.com/api/v1/payloads/ingest \
-H 'X-API-Key: asig_live_xxxx' -H 'Idempotency-Key: <uuid4>' \
-H 'X-HMAC-Signature: <hex>' -H 'X-Timestamp: 1749461234' \
-H 'Content-Type: application/json' \
-d '{"transaction_reference":"HIPC-001","tx_hash":"0xabc...",
"sender_wallet":"0xYOUR","receiver_wallet":"0xBD682c...",
"amount":"50000","asset":"USDT","network":"ethereum",
"token_contract":"0xdAC17F...", "callback_url":"https://your/webhook"}'
```

13.3 Check Payload Status

```
curl https://api.alshumookh.com/api/v1/payloads/<payload_id> \
-H 'X-API-Key: asig_live_xxxx'
```

13.4 Get Schema

```
curl https://api.alshumookh.com/api/v1/payloads/schema \
-H 'X-API-Key: asig_live_xxxx'
```

13.5 Submit OAuth2 Bearer Token Payload

```
curl -X POST https://api.alshumookh.com/api/v1/payloads/ingest \
-H 'Authorization: Bearer eyJhbGci...' \
-H 'Idempotency-Key: <uuid4>' -H 'Content-Type: application/json' \
-d '{"txHash":"0xabc...", "fromAddress":"0xYOUR", "toAddress":"0xBD682c...",
"tokenAmount":"50000", "tokenSymbol":"USDT", "chainName":"ethereum"}'
```

14. GO-LIVE COMPLIANCE CHECKLIST

ALL items must be completed and verified by ALSHUMOOKH GROUP before HIPERCAPITAL FINANCE receives production API access. Items marked BLOCKER prevent go-live regardless of other progress.

#	Category	Requirement	Evidence Required	Priority
1	Legal	FCA Authorization or equivalent financial license	License cert + regulator register entry	BLOCKER
2	Legal	AML/KYC program documentation	AML policy + compliance officer contact	BLOCKER
3	Legal	Corporate structure + beneficial ownership	Entity chart + UBO declaration	Required
4	Financial	Independent reserve audit (6.98T EUR claim)	Big 4 or equivalent audit report	BLOCKER
5	Financial	HCEURO → USDT bridge on public chain	Bridge contract + audit + liquidity proof	BLOCKER
6	Technical	10 successful Sepolia test payloads	ASIG payload IDs with RECONCILED status	Required
7	Technical	HMAC-SHA256 signing verified	Demo request with valid HMAC header	Required
8	Technical	Webhook endpoint tested	All status transitions received OK	Required
9	Technical	Idempotency-Key implementation verified	Duplicate request returns replay response	Required
10	Technical	Rate limit compliance (200 req/60s)	Load test results — no violations	Required
11	Ops	Production IP allowlisted in ASIG	Egress IPs provided to ASIG ops	Required
12	Ops	Integration SLA signed	Signed SLA document	Required

■ ~~BLOCKERS~~ items (1,2,4,5) must be resolved before any production access is granted. The USDT bridge (#5) is the primary technical blocker — HCEURO has no external liquidity and ASIG cannot accept private ledger assets under any circumstances.